START programme · multiuser XR neuromodulation

mindX

Neurofeedback for two

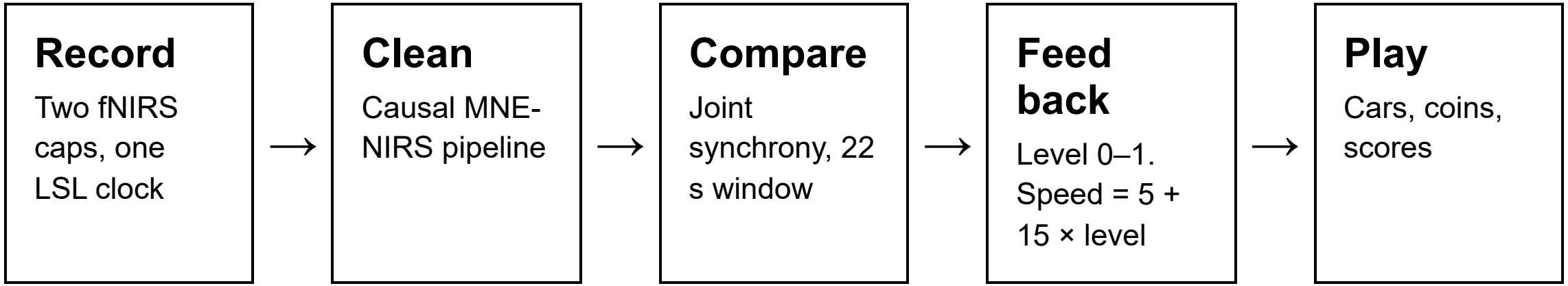
Two brains, one shared signal, fed back as speed in a cooperative race.

Demo with synthetic data · 15 minutes

Can two people learn to synchronise their brains?

Does it carry over to cooperation?

From two heads to one car



Causal and soft real-time. About 7 ms of processing per frame.

fNIRS is slow

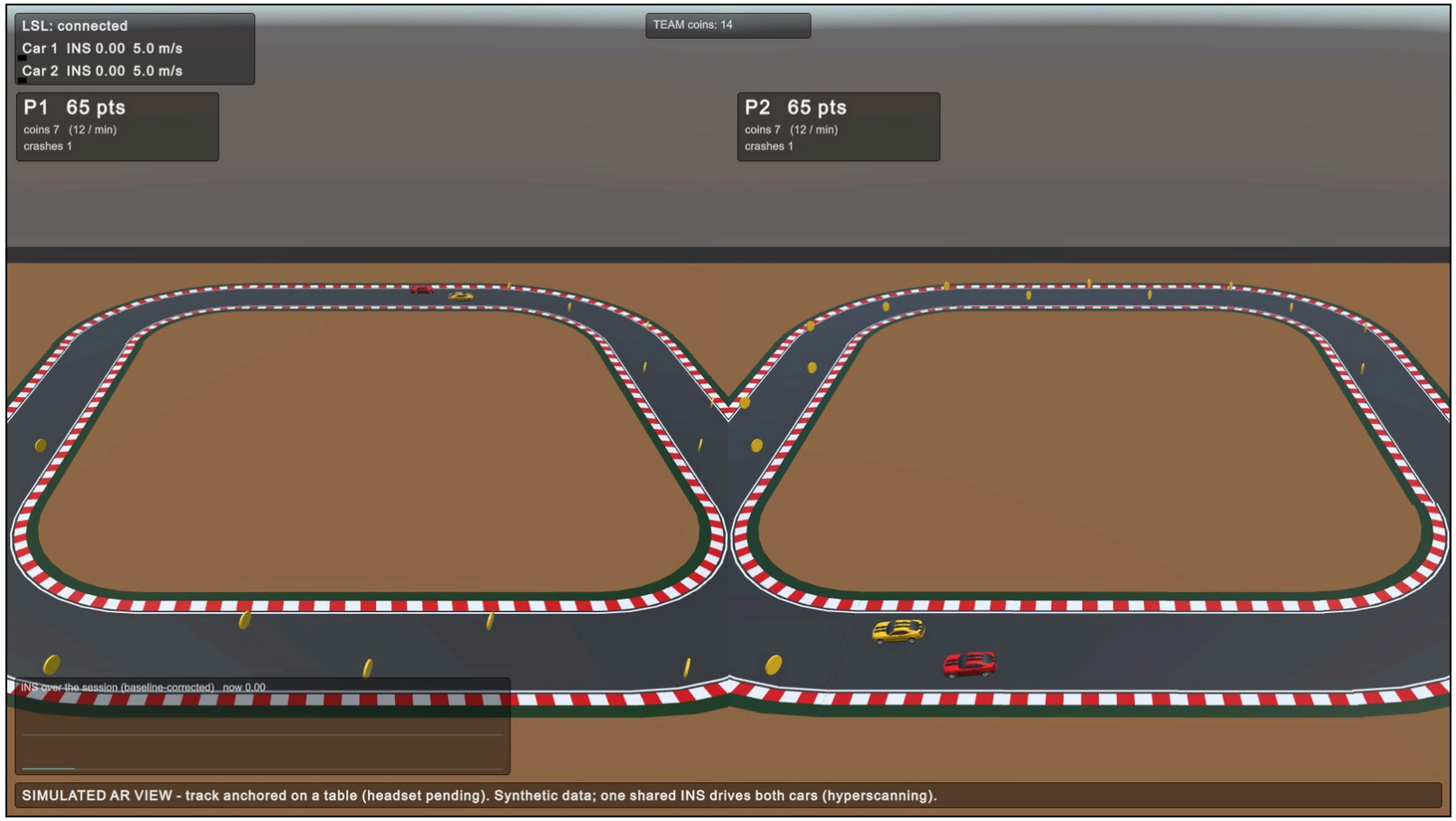
5–7 s
haemodynamic lag

22 s
window the estimate needs

~30 s
until the car reacts

Physiology, not a bug. The demo ramp is sped up.

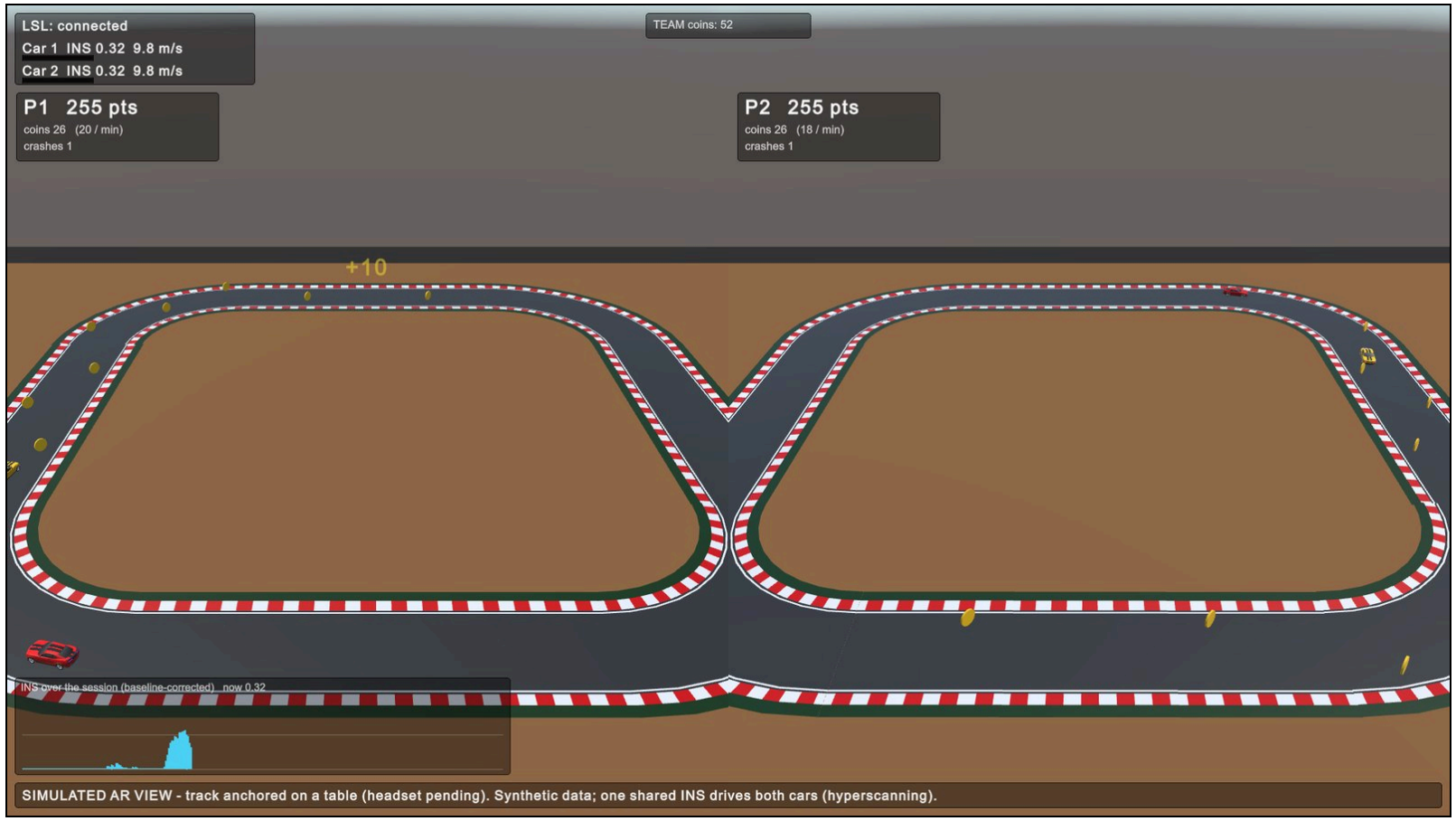
Baseline: level zero



- Level 0, speed 5 m/s
- One view per player
- Own score in each view
- 10 points per coin

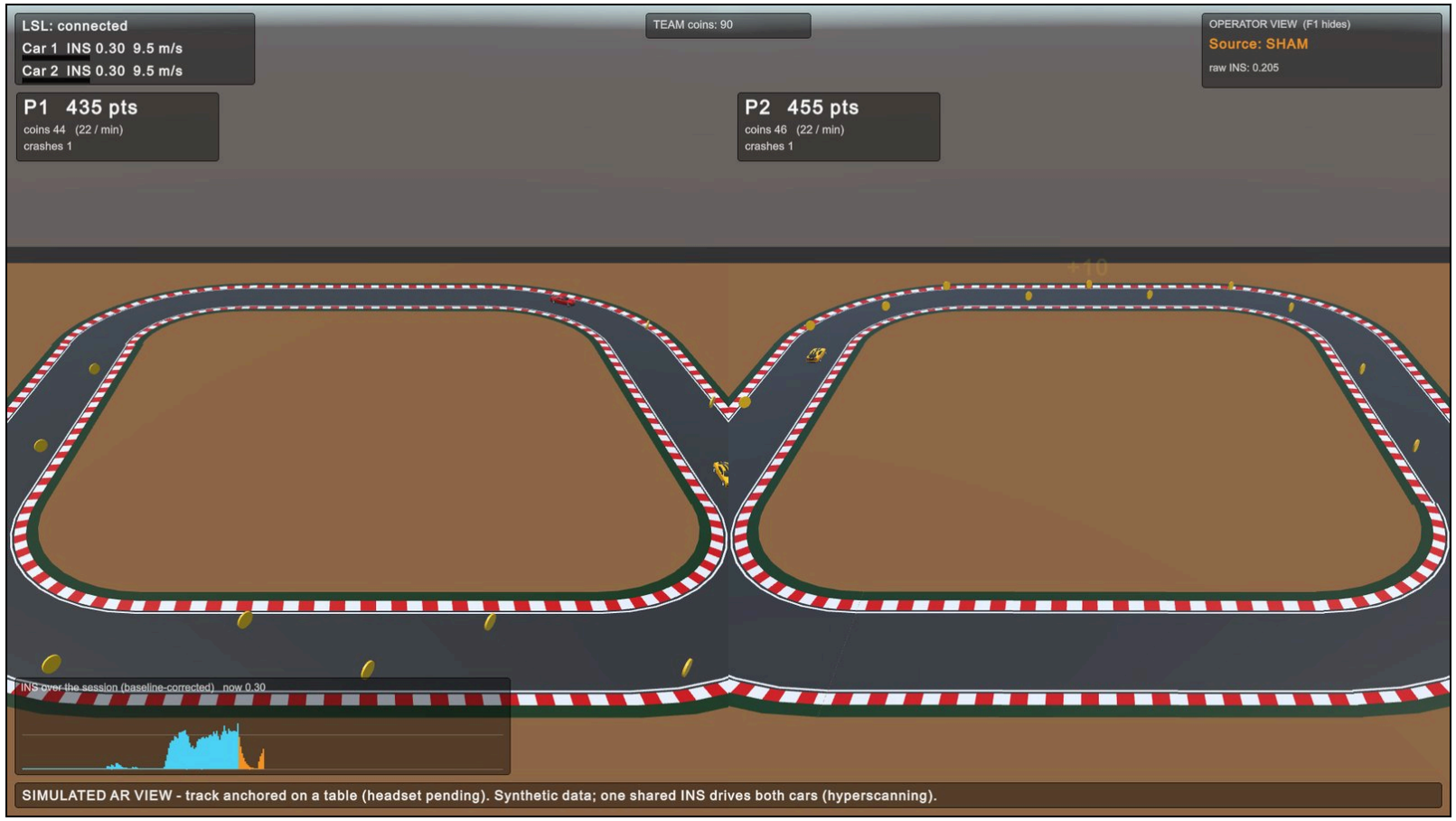
Synthetic data · screenshot of the live session

The signal rises, the car follows



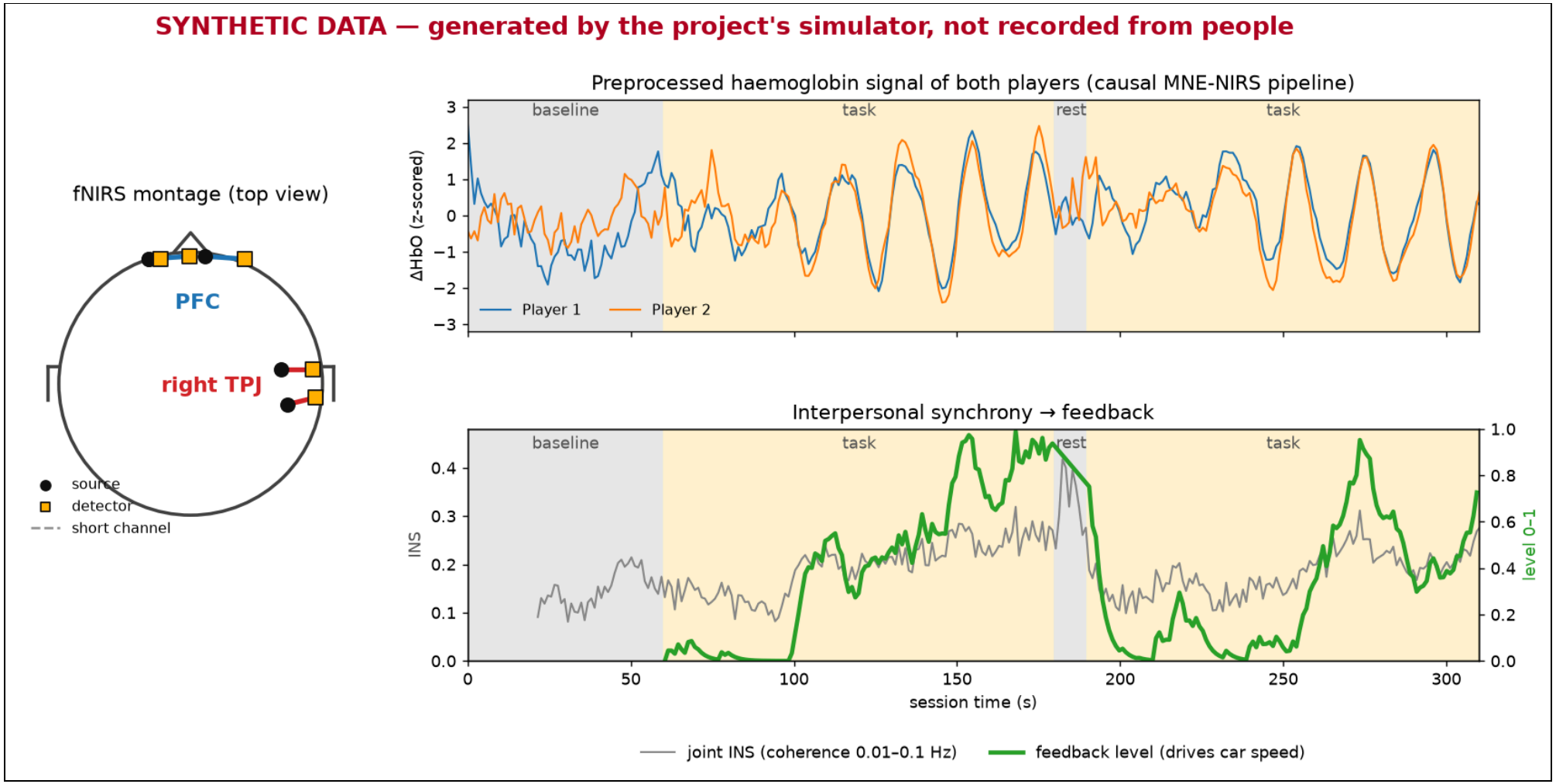
- Speed 5 to 17 m/s
- Coins per minute: 11 to 27
- Delay about 30 s
- TEAM total on top

Sham: another dyad's recording

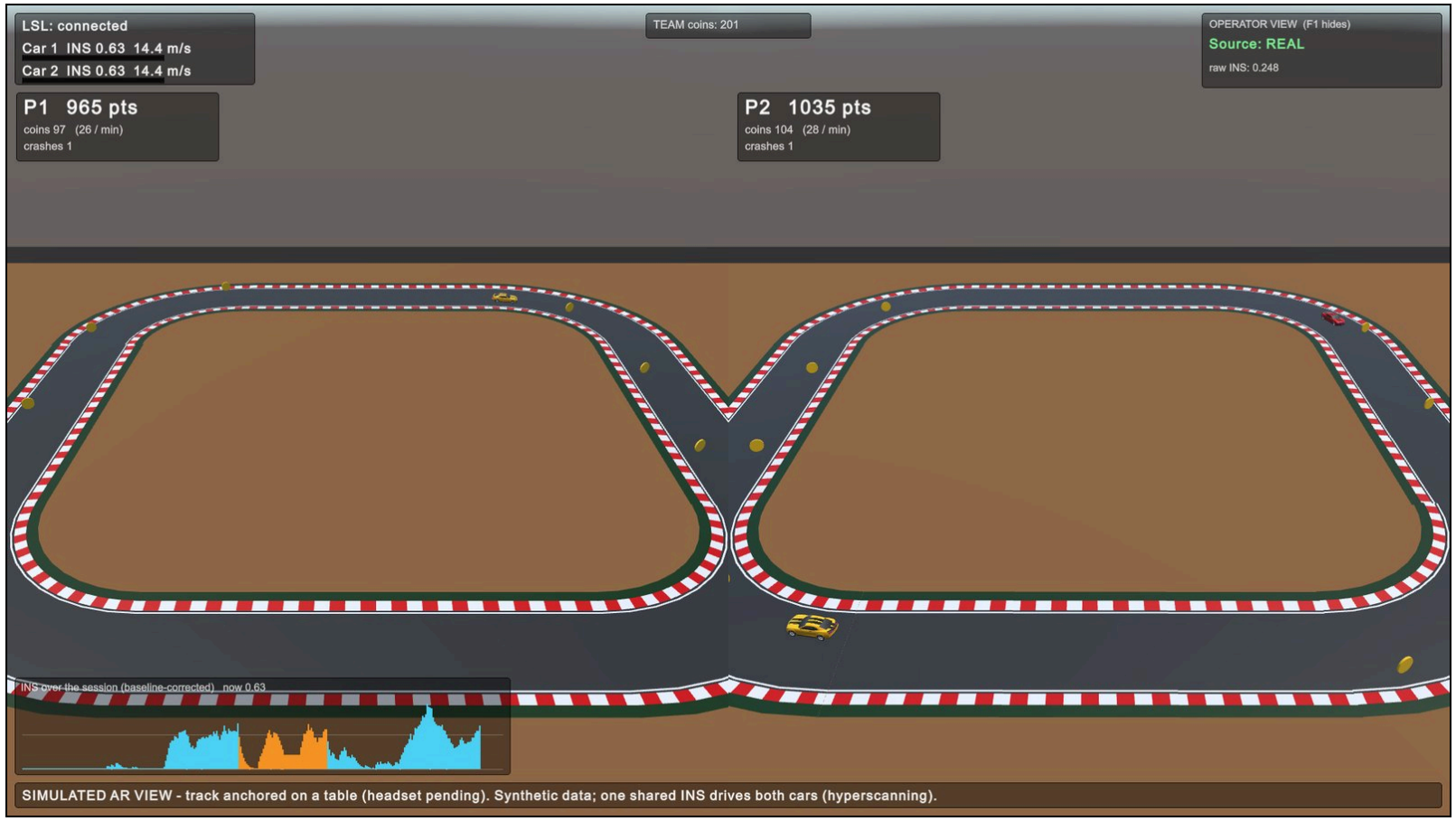


- Same mapping, other source
- Players cannot tell
- Orange: operator view only
- It also rises: real trajectory

What the sensors see (synthetic)



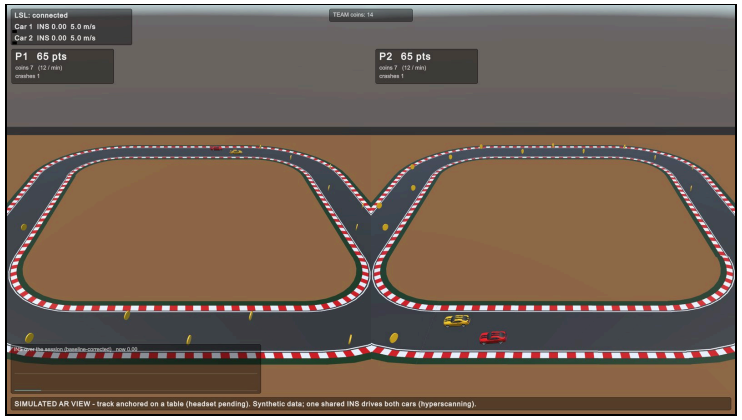
Two players, two scores



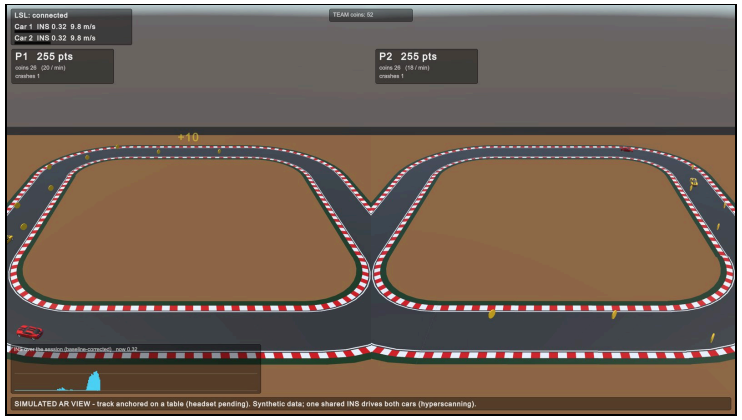
- +10 per coin, own lane
- −5 each on a crash
- Scores differ by about 7%
- Same signal, same speed

Synthetic data · screenshot of the live session

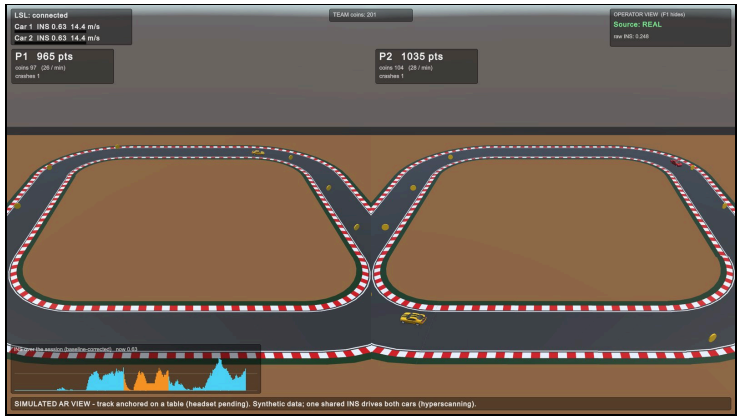
The whole session in 19 seconds



baseline



signal rising



second peak

- 5 minutes of play, shown at about 18× speed
- Video file: session_timelapse.mp4
- Same clip as a GIF: session_timelapse.gif

AR on the table, simulated

Game world

Full size, in metres. Physics, coins and scores run here.

→

Table copy

Scaled by 0.0075. Visual only. One view per player.

Passthrough and anchoring arrive with the headset.

Real code, synthetic data

Synthetic

- fNIRS of both players
- Shared 0.05 Hz component
- The sham partner
- The AR table view

Real

- MNE-NIRS online pipeline
- Causal coherence estimate
- LSL into the game
- Coins, scores, penalty
- Automated tests

Next steps

Level B

One person, live fNIRS

2–4 weeks. Needs hardware.

Level C

Two people, live

1–2+ months.

Open

Per-player feature (O7)
XR rig on XRI 3
Varjo XR-4

Questions?